

HDOC Day-17

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SUJECT: C programming

DATE: 10-09-2026

1. Problem Statement

Write a menu-driven C program that performs digit-based operations on a positive integer. The menu must use switch-case. Each menu option should call a separate user-defined function with void return type and no arguments. The selected function itself reads the integer, processes it, and displays the result.

- Generate the largest possible number using the given digits.
- Generate the smallest possible number using the given digits.
- Swap the first and last digits.
- Check whether the number is a palindrome.

2. Algorithm

1. Start the program.
2. Display a menu containing the four digit operations.
3. Read the user's menu choice.
4. Use switch-case to call the function related to the selected option.
5. For the largest number, count how many times each digit occurs and print the digits from 9 down to 0.
6. For the smallest number, count the digits, print the smallest non-zero digit first, then all zeros, followed by the remaining digits in ascending order. This avoids a leading zero.
7. For swapping, find the first digit, last digit and place value of the first digit, then rebuild the number after interchanging the two end digits.
8. For palindrome checking, reverse the number and compare the reversed value with the original number.
9. Display the result produced by the selected function.
10. Stop.

3. Test Case Table

Choice	Operation	Input	Expected Output	Purpose
1	Largest number	31052	53210	Digits arranged in descending order
2	Smallest number	31052	10235	Smallest non-zero digit is kept first
2	Smallest number	90010	10009	Checks correct handling of zeros
3	Swap first & last	12345	52341	First and last digits are interchanged
3	Swap first & last	7	7	Single-digit number remains unchanged
4	Palindrome check	12321	Palindrome	Same when read forward and backward

Choice	Operation	Input	Expected Output	Purpose
4	Palindrome check	12345	Not palindrome	Reverse is different from original
9	Invalid menu choice	-	Invalid choice	Checks default case

4. C Program

```
#include <stdio.h>

void largestNumber(void);
void smallestNumber(void);
void swapFirstLast(void);
void checkPalindrome(void);

int main(void)
{
    int choice;

    printf("Digit Operations Menu\n");
    printf("1. Generate largest possible number\n");
    printf("2. Generate smallest possible number\n");
    printf("3. Swap first and last digits\n");
    printf("4. Check palindrome\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            largestNumber();
            break;
        case 2:
            smallestNumber();
            break;
        case 3:
            swapFirstLast();
            break;
        case 4:
            checkPalindrome();
            break;
        default:
            printf("Invalid choice.\n");
    }

    return 0;
}

void largestNumber(void)
{
    int n;
    int count[10] = {0}, digit, i;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while (n > 0)
    {
        digit = n % 10;
```

```

        count[digit]++;
        n /= 10;
    }

    printf("Largest possible number: ");
    for (i = 9; i >= 0; i--)
        while (count[i]-- > 0)
            printf("%d", i);
    printf("\n");
}

void smallestNumber(void)
{
    int n;
    int count[10] = {0}, digit, i, first = -1;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    while (n > 0)
    {
        digit = n % 10;
        count[digit]++;
        n /= 10;
    }

    for (i = 1; i <= 9; i++)
    {
        if (count[i] > 0)
        {
            first = i;
            count[i]--;
            break;
        }
    }

    printf("Smallest possible number: ");
    if (first != -1)
        printf("%d", first);

    while (count[0]-- > 0)
        printf("0");

    for (i = 1; i <= 9; i++)
        while (count[i]-- > 0)
            printf("%d", i);
    printf("\n");
}

void swapFirstLast(void)
{
    int n, temp, divisor = 1, middle, result;
    int first, last;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n < 10)
    {
        printf("After swapping: %d\n", n);
        return;
    }

```

```

    }

    last = n % 10;
    temp = n;
    while (temp >= 10)
    {
        temp /= 10;
        divisor *= 10;
    }
    first = temp;

    middle = (n % divisor) / 10;
    result = last * divisor + middle * 10 + first;

    printf("After swapping: %d\n", result);
}

void checkPalindrome(void)
{
    int n, temp, reverse = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    temp = n;
    while (temp > 0)
    {
        reverse = reverse * 10 + temp % 10;
        temp /= 10;
    }

    if (reverse == n)
        printf("%d is a palindrome number.\n", n);
    else
        printf("%d is not a palindrome number.\n", n);
}

```

5. Writing and Compilation

The program can be saved as day17.c. In a Linux terminal, compile it with GCC using the command below. The program was compiled with warnings enabled, and the compilation completed successfully.

```
gcc -Wall -Wextra -std=c11 day17.c -o day17
```

Compilation status: Successful (no compiler warnings/errors)

6. Execution

Run the executable with the following command. The menu appears first, then the chosen function asks for the required positive integer.

```
./day17
```

7. Testing Against the Prepared Cases

Test 1 - Largest possible number

```
Terminal - Kali Linux

yug@kali:~/HDOC$ ./day17
Digit Operations Menu
1. Generate largest possible number
2. Generate smallest possible number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 1
Enter a positive integer: 31052
Largest possible number: 53210
yug@kali:~/HDOC$
```

Test 2 - Smallest possible number

```
Terminal - Kali Linux

yug@kali:~/HDOC$ ./day17
Digit Operations Menu
1. Generate largest possible number
2. Generate smallest possible number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 2
Enter a positive integer: 31052
Smallest possible number: 10235
yug@kali:~/HDOC$
```

Test 3 - Smallest number with repeated zeros

```
Terminal - Kali Linux

yug@kali:~/HDOC$ ./day17
Digit Operations Menu
1. Generate largest possible number
2. Generate smallest possible number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 2
Enter a positive integer: 90010
Smallest possible number: 10009
yug@kali:~/HDOC$
```

Test 4 - Swap first and last digits

```
Terminal - Kali Linux

yug@kali:~/HDOC$ ./day17
Digit Operations Menu
1. Generate largest possible number
2. Generate smallest possible number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 3
Enter a positive integer: 12345
After swapping: 52341
yug@kali:~/HDOC$
```

Test 5 - Palindrome

```
Terminal - Kali Linux

yug@kali:~/HDOC$ ./day17
Digit Operations Menu
1. Generate largest possible number
2. Generate smallest possible number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 4
Enter a positive integer: 12321
12321 is a palindrome number.
yug@kali:~/HDOC$
```

Test 6 - Not a palindrome

```
Terminal - Kali Linux

yug@kali:~/HDOC$ ./day17
Digit Operations Menu
1. Generate largest possible number
2. Generate smallest possible number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 4
Enter a positive integer: 12345
12345 is not a palindrome number.
yug@kali:~/HDOC$
```

8. Result

The menu-driven C program was written, compiled and executed successfully. All four operations produced the expected results for the prepared test cases. The switch-case menu correctly calls a separate void function for each option, and each function independently reads its input, performs the required digit operation and displays the result.